

Exact Organization of Density Matrices and Entanglement Structure in the Kitaev Spin Liquid

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1 Kitaev Model

Kitaev Model: an Introduction

- Model Hamiltonian

$$H = - \sum_{\langle ij \rangle} J_{\alpha_{ij}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} \quad ; \quad \begin{array}{c} \text{Diagram of two adjacent hexagons sharing a vertical edge. The top vertex of the right hexagon is labeled } x, \text{ the top vertex of the left hexagon is labeled } y, \text{ and the shared vertical edge is labeled } z. \end{array}$$

- Majorana fermion (parton) representation: $b^x, b^y, b^z, c \Rightarrow \sigma^\mu = i b^\mu c$
- Parton Hamiltonian: define $i b_i^{\alpha_{ij}} b_j^{\alpha_{ij}} = \hat{u}_{ij}$

$$H \rightarrow \sum_{\langle ij \rangle} i J_{\alpha_{ij}} \hat{u}_{ij} c_i c_j$$

- Gutzwiller projection

$$\mathcal{P}_G = \prod_i \left(\frac{1 + D_i}{2} \right) \quad , \quad D_i = b_i^x b_i^y b_i^z c_i \Leftarrow -i \sigma_i^x \sigma_i^y \sigma_i^z$$

- Separability of the entanglement entropy (Yao and Qi, PRL **105** 080501)

$$S_A = S_A^F + S_A^G$$

Kitaev Model: Gauge Theory

In the Majorana fermion representation, the model is equivalent to a system of free Majorana fermions (matter field) coupled to a static $\mathbb{Z}_2$ gauge field

- Define the gauge field $\hat{u}_{ij} = e^{iA_{ij}}$, then the elementary $\mathbb{Z}_2$ gauge flux reads

$$e^{i\Phi} = \exp\left(i \oint_{\partial p} A\right) = \prod_{\langle ij \rangle \in \partial p} \hat{u}_{ij} = \prod_{\langle ij \rangle \in \partial p} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} =: W_p$$

- The electric field in the gauge field that flips the gauge field is $b_i^{\alpha_{ij}} \rightarrow e^{i\pi E_{ij}}$ because $b_i^{\alpha_{ij}} \hat{u}_{ij} b_i^{\alpha_{ij}} = -\hat{u}_{ij}$
- Gutzwiller projection as a Gauss law constraint: $D_i = b_i^x b_i^y b_i^z c_i = 1$. The ground state is

$$\prod_i \left(\frac{1 + D_i}{2} \right) |u\rangle |M\rangle$$

- Both definitions $b_i^{\alpha_{ij}} = e^{i\pi E_{ij}}$ and $b_j^{\alpha_{ij}} = e^{i\pi E_{ij}}$ are valid.
- The Gauss law is a physical constraint and cannot be broken.

Kitaev Model: Properties of Spin Correlation Functions

- Two-spin correlation functions are short-ranged (Baskaran, Mandal, and Shankar, PRL**98**, 247201)